

EXPERIMENT - 5

Date: 2082/09/02

TO DETERMINE THE COEFFICIENT OF LINEAR EXPANSION OF THE GIVEN ROD BY USING PULLINGER'S APPARATUS

APPARATUS REQUIRED:

- Pullinger's apparatus
- Thermometer
- Spherometer
- Heating arrangement & boiler
- Meter scale
- Galvanometer

THEORY:

The coefficient of linear expansion of a solid in the form of a tube or a rod is the increase in its length per unit length per unit degree rise of temperature. Let,

Initial length of the rod at initial temperature θ_1 °C = l_1

Final length of the rod at final temperature θ_2 °C = l_2

Now, the increase in length for a rise of temperature from θ_1 °C to θ_2 °C = $l_2 - l_1$

Then, coefficient of linear expansion (α) = $\frac{\text{Increase in length}}{\text{Original length} \times \text{Rise in Temperature}}$

$$\text{i.e. } \alpha = \frac{l_2 - l_1}{l_1 \times (\theta_2 - \theta_1)} = \frac{\Delta l}{l_1 \times (\theta_2 - \theta_1)}$$

OBSERVATION:

Material of the rod supplied = Brass

Least count of meter scale = 0.1 cm

Original length of the rod (l_1) = 5.2 cm

Pitch of the spherometer (P) = 1 mm

No. of circular scale divisions (N) = 100 divs.

Then, least count of the spherometer (L.C.) = $\frac{\text{Pitch}}{\text{No. of circular scale division}}$

$$\text{i.e. } \text{L.C.} = \frac{P}{N} = \frac{1}{100} = 0.01 \text{ mm}$$

$$\text{or, } \text{L.C.} = 0.001 \text{ cm}$$

Initial temperature of the rod (θ_1) = 10 °C

Initial circular scale reading on the upper surface (a) = 50 div.

No. of complete rotations through which the circular scale is raised (X) = 2

No. of complete rotations lowered (Y) = 1

Final circular scale reading on the upper surface (b) = 5 div.

Final steady temperature of the rod (θ_2) = 80 °C

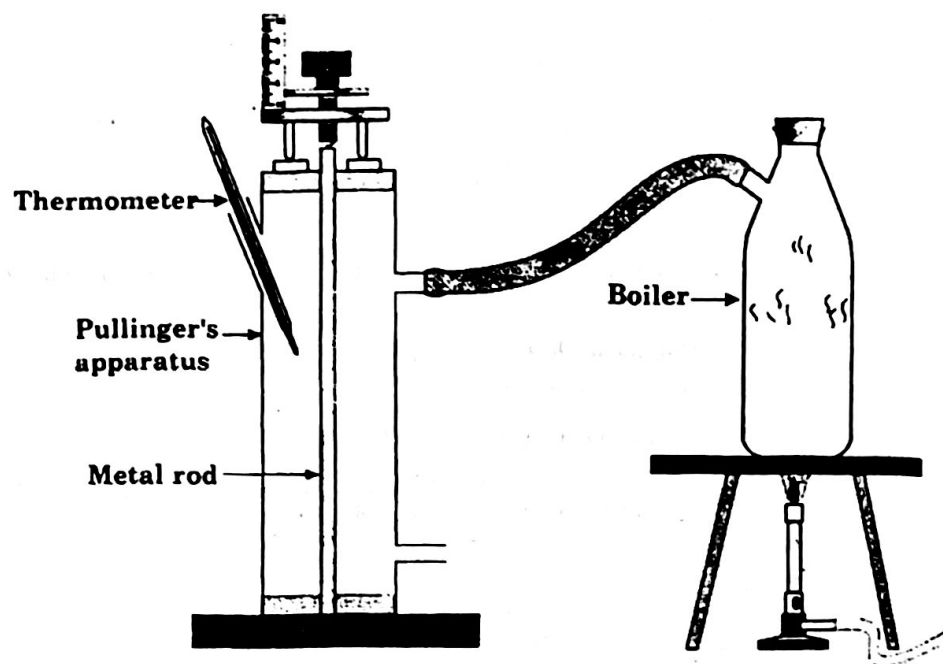


Fig. Pullinger's apparatus

CALCULATION:

$\Delta l = X \times P - [Y \times P + (a - b) \times \text{L.C.}]$ for $a > b$ [but if $a < b$ then $\Delta l = X \times P - [Y \times P + (100 + a - b) \times \text{L.C.}]$]

$$= 2 \times 1 - [1 \times 1 + (40 - 5) \times 0.01]$$

$$= 0.55 \text{ mm}$$

$$= 0.055 \text{ cm}$$

$$\alpha = \frac{l_2 - l_1}{l_1 \times (\theta_2 - \theta_1)} = \frac{\Delta l}{l_1 \times (\theta_2 - \theta_1)} = \frac{0.055}{40 \times (80 - 10)^\circ\text{C}}$$

Now

$$\alpha = 19.6 \times 10^{-6} / ^\circ\text{C}$$

RESULT:

The value of linear expansivity of material of given rod (α) = $19.6 \times 10^{-6} / ^\circ\text{C}$.

PERCENTAGE ERROR:

For Brass.....,

Standard value (S.V.) of linear expansivity, $\alpha = 18.9 \times 10^{-6} / ^\circ\text{C}$

$$\% \text{ error} = \left| \frac{\text{S.V.} - \text{O.V.}}{\text{S.V.}} \right| \times 100\% = \frac{18.9 \times 10^{-6} - 19.6 \times 10^{-6}}{18.9 \times 10^{-6}} \times 100\% = 3.70\%$$

CONCLUSION:

Hence, from above experiment, the linear expansivity of the material of given rod was determined using Pullinger's apparatus.

SOURCES OF ERROR:

The error may arise due to

- ① Touching or pressing hard against the rod by the central leg of spherometer.
- ② Backlash error of the spherometer.
- ③ Carelessness in the experiment.

PRECAUTIONS:

- ① The central leg of spherometer must touch the rod gently.
- ② The spherometer should be twisted only in one direction for an observation.
- ③ The readings should be taken carefully.

VIVA QUESTIONS:

1. What is coefficient of linear expansion? Does it depend upon length?
2. What is differential expansion?
3. Why does a solid expand on heating?
4. Pendulum clocks generally run slow in summer and faster in winter. Why?
5. Why does a beaker filled with water at 4°C over-flow if the temperature is decreased or increased?
6. What do you mean by cubical superficial expansivity?
7. Why are the standard pendulum clock made of invar?
8. What do you mean by the linear expansivity of brass is $19 \times 10^{-6} \text{ }^{\circ}\text{C}^{-1}$?

① Coefficient of linear expansion is defined as the change in length per unit original length per unit change in temperature. It is denoted by α . If l_1 and l_2 are the lengths of the rod at temperatures θ_1 and θ_2 respectively then

$$\alpha = \frac{l_2 - l_1}{l_1 (\theta_2 - \theta_1)}$$

It doesn't depend upon length, it depends upon the nature of the material.

②

② Differential expansion is the difference in expansion shown by two different metals when the same amount of heat is given to them for the same time.

③ As the Temperature increases, the intermolecular separation between molecules increases. So, solid expands on heating.

④ Pendulum clocks run faster in winter and slow in summer because of thermal expansion and

Contraction.

In summer, the pendulum rod expands, its length increases, so the time period increases and the clock runs slow.

In winter, the rod contracts, its length decreases, so the time period decreases and the clock runs fast.

⑤ Due to anomalous expansion of water, water has minimum volume at 4°C . Therefore water expands whether it is heated above 4°C or cooled below 4°C .

⑥ Cubical expansivity is the increase in volume per unit original volume of a substance for a rise of 1°C in temperature, while Superficial expansivity is the increase in area per unit original area of a substance for a rise of 1°C in temperature.

⑦ Pendulum clocks are made of Invar because it has a very low coefficient of thermal expansion so, it hardly expands with temperature, keeping the clock accurate.

⑧ It means that 1m length of a brass rod will increase by $19 \times 10^{-6}\text{m}$ when it is heated through the temperature of 1°C .

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